VINUNIVERSITY

Spring 2025

COMP1020: Object-Oriented Programming & Data Structures

Lecture 1: Course Introduction

Course Organization, Course Contents and Basic Concepts in OOP and Java

Slides developed by Hieu Pham and Kok-Seng Wong, College of Engineering & Computer Science, VinUniversity

February 18^{th}, 2025

Outline: Plan for today

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- ✓ Course Organization and Logistics
- ✓ Learning Objectives and Expected Learning Outcomes
- ✓ Introduction to Object-Oriented Programming (OPP)
- ✓ Introduction to Java

TEACHING TEAM

ABOUT THE INSTRUCTORS

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About me (Lecture 1 to Lecture 6)

- Ph.D. in Computer Science, Univ. of Toulouse, France (2019).
- Research Scientist and Head of Fundamental Research at Medical Imaging Center, Vingroup Big Data Institute (2019 - 2021)
- Assistant Professor, CECS, VinUniversity (present)
- Principal Investigator (PI), VinUni-Illinois Smart Health Center, VinUniversity. Head of Computer Vision and Medical AI Lab (present)
- Scientific Director of Entrepreneurship Lab (E-lab) @ VinUni (present)
- Research Interests: Computer Vision, Artificial Intelligence, Machine Learning, Deep Learning, Medical Image Analysis, and Smart Healthcare.

Academic webpage: https://huyhieupham.github.io/
Google Scholar: https://shorturl.at/wxzJ8
Email: hieu.ph@vinuni.edu.vn

Scan to learn more about our research

ABOUT THE INSTRUCTORS

Professor Sungdeok Cha (From Lecture 7 onward)

- Ph.D., Information and Computer Science, University of California, Irvine, 1991
- Member of Technical Staff, The Aerospace Corporation, USA (1991~1994)
- Professor, Computer Science Department, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea (1994~2008)
- Professor, Computer Science and Engineering, Korea University, Seoul, Korea (2008~2025)
- Professor Emeritus, Computer Science and Engineering, Korea University, Seoul, Korea (2025~)
- Professor, CECS, VinUniversity (March 2025~)
- Research Interest: Software Engineering, Requirements Engineering, Software Safety, Computer Security
- Academic webpage: http://dependable.korea.ac.kr/
- Google Scholar: https://shorturl.at/4sjMZ
- Email: scha@korea.ac.kr

Professor Sungdeok Cha

TEACHING ASSISTANTS

Le Minh Huyen

B.Eng. in Applied Information Engineering from Yonsei University in 2023. Now CS Ph.D. student on AI in Biomedical Imaging at VinUniversity.

Pham Quoc Cuong

B.Eng. from VNU-HCM University of Technology and M.Eng. from Ritsumeikan University, Japan. Now CS Ph.D. studenton AI in Healthcare at VinUniversity.

Our TAs will be responsible for lecture support, assisting with assignments and projects, holding office hours, providing other student support, and conducting weekly lab sessions. More on Canvas.

COURSE ORGANIZATION

Course Information, Learning Objectives and Expected Learning Outcomes

CLASS SCHEDULE

Two lectures and two laboratory sessions per week

❖ **Lecture**

Group	Day & Time	Venue
All students	Tuesdays: 11.00 am – 12.15 pm	E402
	Thursdays: 11.00 pm – 12.15 pm	E402

❖ **Laboratory**

Group	Day & Time	Venue
Session 1	Mondays: 1.10 pm – 3.00 pm	G102 (Super Lab)
Session 2	Fridays: 10.30 pm – 12.20 pm	G102 (Super Lab)

During lectures, students are encouraged to ask questions and interact with the Instructor. During laboratories, students engage in program development and discussion with instructors who facilitate the laboratory work.

COURSE DESCRIPTION

- **Course Information**: This is a required 4-credit course.
- **Prerequisite**: COMP1010 Introduction to Programming.
- **Level**: This course is an intermediate programming course taught to all CS and DS students.
- **Rationale:** This course extends the basic programming course by providing students with an understanding of object-oriented programming languages and skills in object-oriented programming.

LEARNING OBJECTIVES

Upon completion of the course, students will be able to:

- Understand and apply the basic concepts of object-oriented programming. Students will understand the philosophy of the OOP methodology. The course will help you understand the differences of the OOP and Procedure Oriented Programming (POP) methodologies.
- Gain hands on experience of programming in Java
- Acquire intermediate programming skills and understand the fundamental algorithms required to engage in more complex computer science courses.
- Develop the ability to solve conceptual and application-based computing problems via regular practice, both in and out of class.

LEARNING OBJECTIVES

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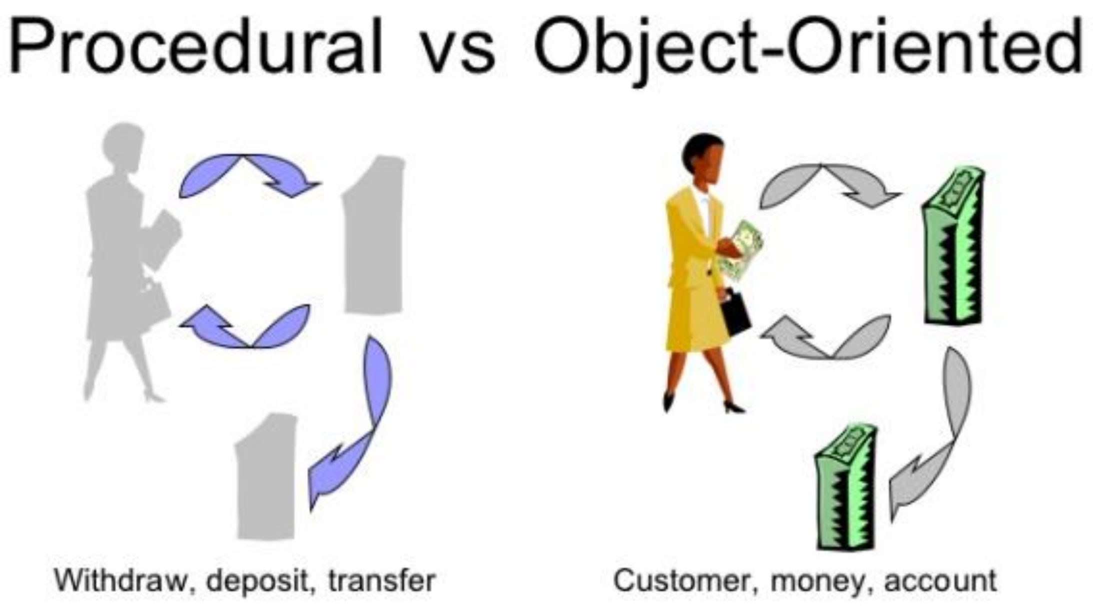
Upon completion of the course, students will be able to:

- Understand why and where you should use OOP instead of POP.
- Start thinking of your programming in terms of “objects” with classes, instead of a traditional “top down” programming approach.
- Write an application or a program using “objects” with classes.



COURSE SYLLABUS

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Week	Topic	Objective
01	Getting started with Java	Get familiarized with a Java program development environment and run first simple programs.
02	Expressions, variables, operators, assignments, statements, and objects in Java.	Acquire the skill of working with the programming language Java basics.
03	Objects, constructors, constructor calls, static members in Java.	Acquire the skill of creating own objects. Be able to create multiple constructors for a class and understand constructor calls in inherited classes.
04	Abstract classes and methods, interfaces, overriding and casting, collections, iterators.	Acquire the skill of implementing abstract classes and methods, interfaces, overriding, and casting.
05	Stacks and queues (data structure).	Acquire the skill of using and creating own stacks and queues in solving computing problems.
06	Basic sorting algorithms and their use in Java.	Acquire the skill of using and creating sorting algorithms in Java.
07	Search trees, balanced trees, and algorithm for tree traversal.	Acquire skills of using and creating search and balanced trees. Be able to create algorithms for tree traversal in Java.
09	Priority queues and heaps (data structure).	Acquire skills of using priority queues and heaps. Be able to understand heapsort in Java.
11	Shortest-path and spanning trees.	Acquire skills of the use and of basic graphs algorithms such as Dijkstra’s shortest-path algorithm.
12	Hashing and its use in Java.	Acquire the skill of using hashing in Java.

COURSE LOGISTICS

- The course will utilize Canvas for managing all learning components, announcements, and course material distribution.
- In laboratories, students use the programming environment of their choice.
- The Eclipse IDE is suggested.
- The lab computer is highly recommended over a personal computer (lab sessions) .
- CMS will be used to evaluate your code solutions. You can also use our CMS for self-learning with test cases. This system is tentatively expected to be available in week 3.

COURSE REQUIREMENTS

Class attendance, participation and etiquette policies:

- Students are required to participate in lectures, and laboratory.
- Students are not allowed to submit quizzes and lab solutions remotely.

COURSE READINGS

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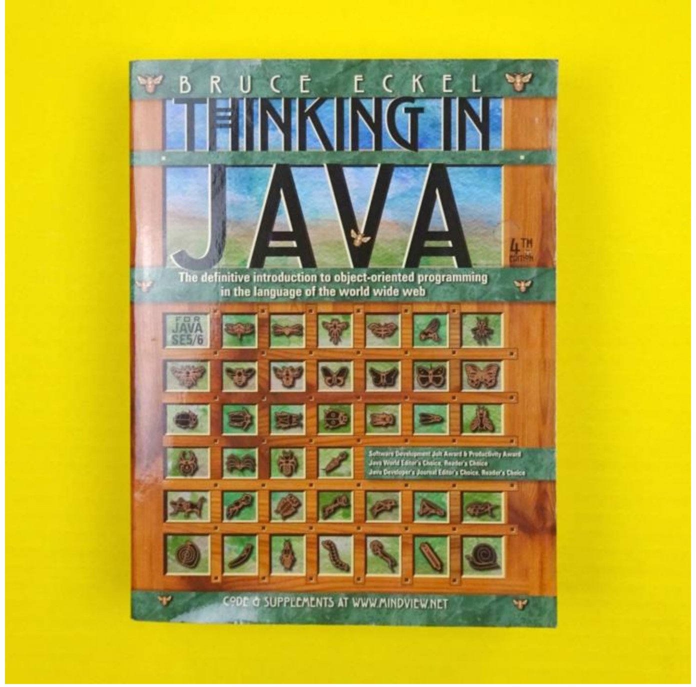
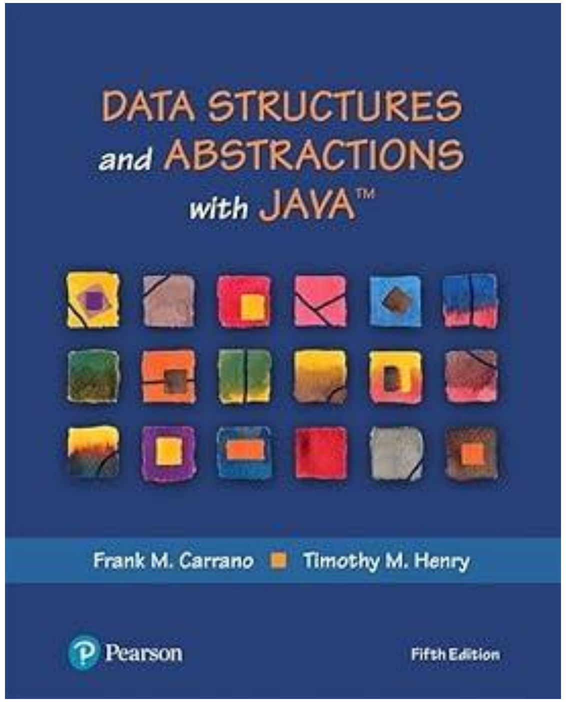
Required readings:

- Eckel, B. (2006) “*Thinking in Java”.* USA: Prentice Hall.
- Carrano, F. M., Henry, T. M. (2014) “*Data Structures and Abstractions with Java”.* USA: Pearson.





GRADING PROCEDURES

GRADING PROCEDURES

Item	Assessment Method	Weighting
1. Class Attendance, Quiz & Presentation	Online Quizzes and Class Presentation	10%
2. Weekly Laboratory	Hands-on Practice	20%
3. Term Project	Group Project (Program, Report, and Presentation)	30%
4. Midterm Exam	Written & Hands-on	20%
5. Final Exam	Written & Hands-on	20%
Total		100%

LABORATORY ASSESSMENT

Weekly Laboratory (20%)

Laboratory	% Weighting
1	-
2	2%
3	2%
4	2%
5	2%
6	2%
7	2%
8	2%
9	2%
10	2%
11	2%
12	2%
Total	Max 20%

Term Project (30%)

Laboratory	% Weighting
Proposal	5%
Reports (Interim and Final)	10%
Program and Presentation	15%
Total	30%

Note 10 labs with the highest scores will be counted as the final total score.

GRADING

GRADING SCHEME (LETTER GRADE)

Name:	Range:	
A	100 %	to 89.5%
A-	< 89.5 %	to 84.5%
B+	< 84.5 %	to 79.5%
B	< 79.5 %	to 74.5%
B-	< 74.5 %	to 69.5%
C+	< 69.5 %	to 64.5%
C	< 64.5 %	to 59.5%
C-	< 59.5 %	to 54.5%
D+	< 54.5 %	to 49.5%
D	< 49.5 %	to 44.5%
D-	< 44.5 %	to 40.0%
F	< 40.0 %	to 0.0%

PROCEDURE ORIENTED PROGRAMMING (POP)

What is POP?

KEY PRINCIPLE OF POP

Procedure-Oriented Programming (POP) is a programming paradigm where a program is structured around a sequence of procedures or functions, focusing on a step-by-step execution of tasks by calling these functions.

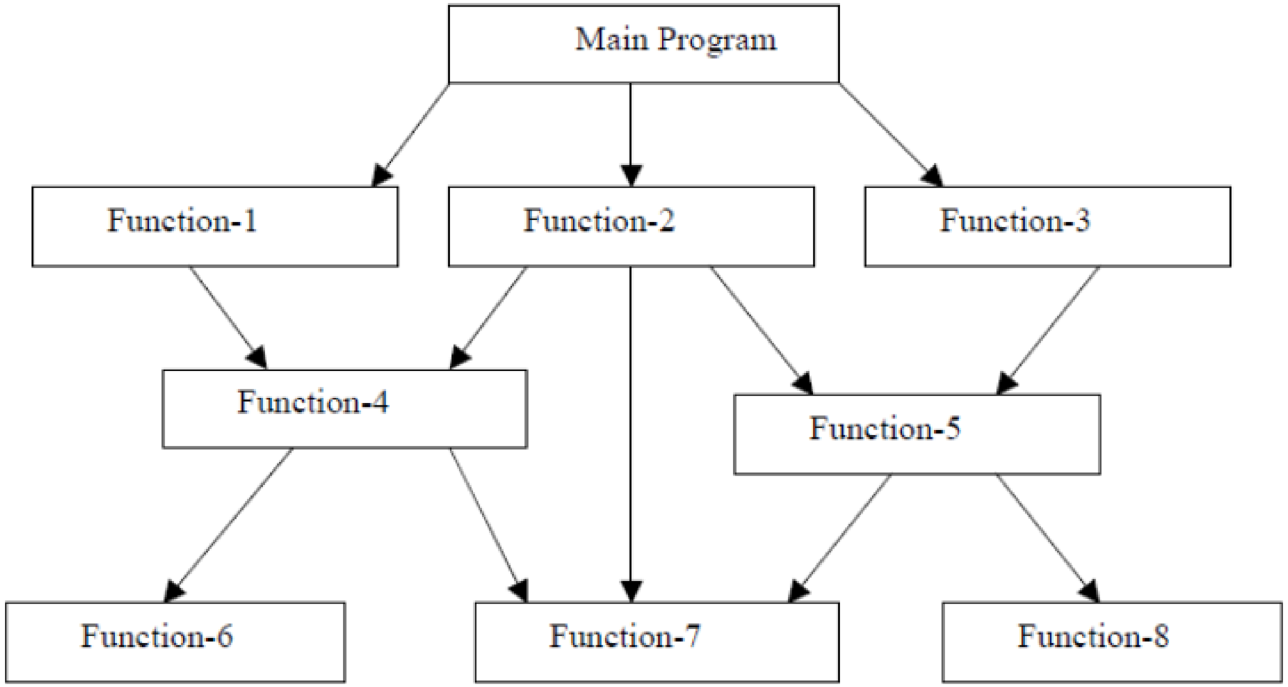
Write a simple program that calculates the sum of two integers and prints the result to the console.

Step 1: Get two numbers.
Step 2: Add these numbers to get the sum.
Step 3: Display the sum to the console.

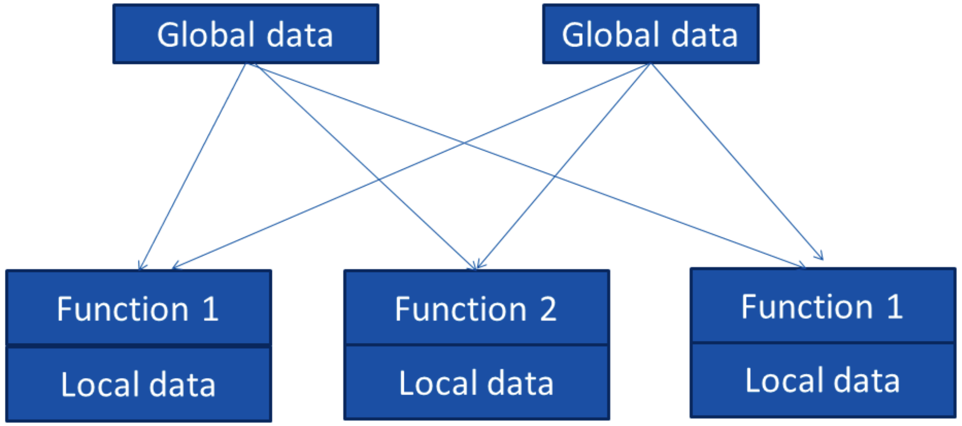
```java
import java.util.*;

public class Main {

    public static void main(String[] args) {

        int x = 89;

        int y = 43;

        int sum = x + y;

        System.out.println("Sum is " + sum);
    }
}
```

LIMITATIONS OF POP

What are the limitations of POP?



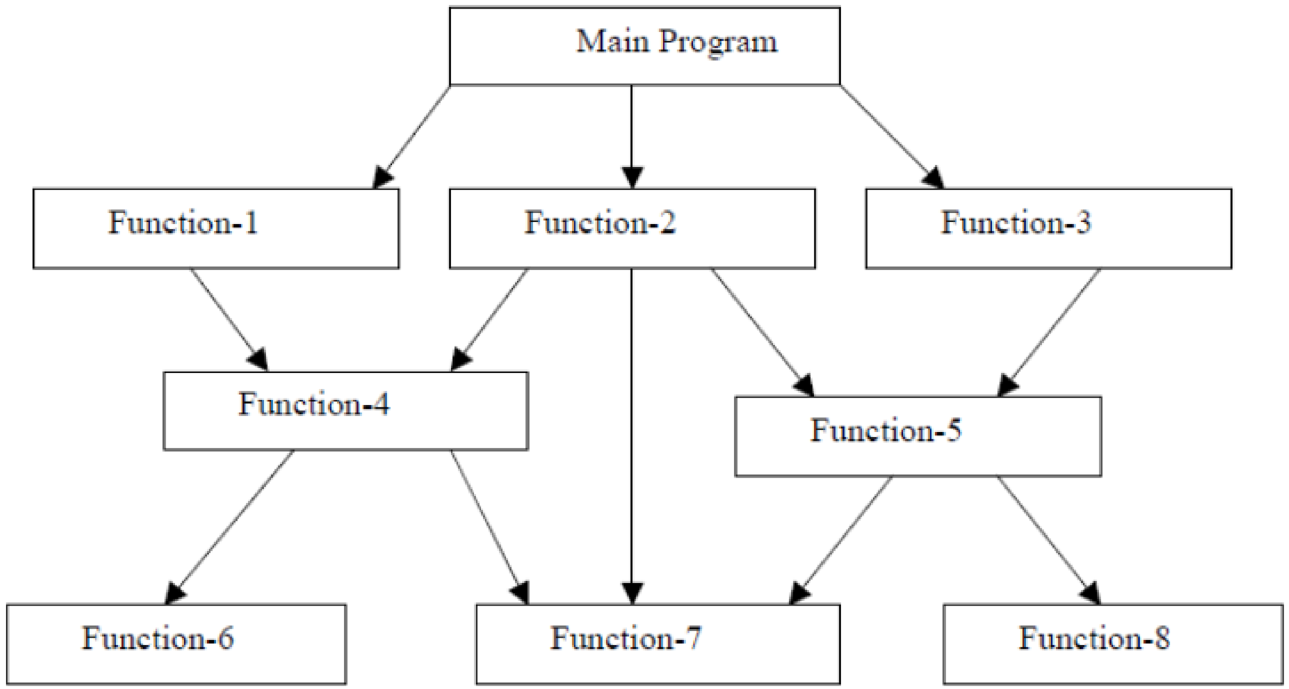
Typical structure of procedural oriented programs



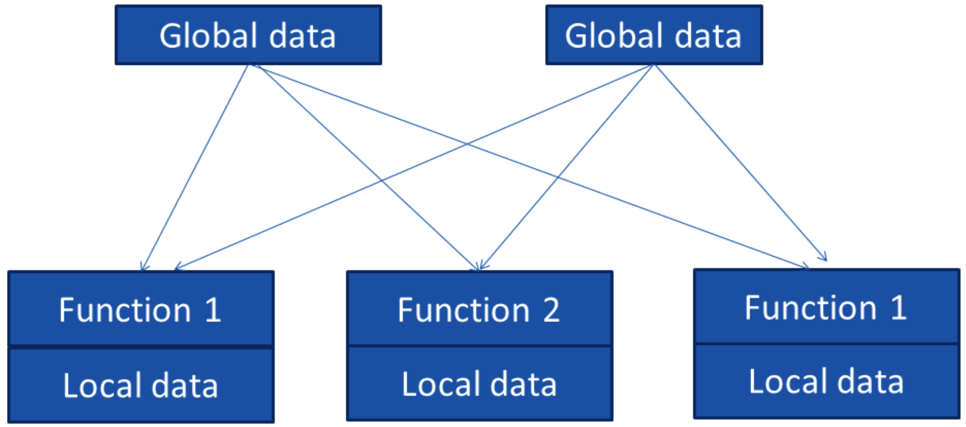
Global variable can be use by any function at any time while local variables are only used within the function.

LIMITATIONS OF POP

It is difficult to scale up and maintain. Also global variables are accessed by all the function so any function can change its value at any time so all the function will be affected.



Typical structure of procedural oriented programs



Global variable can be use by any function at any time while local variables are only used within the function.

INTRODUCTION TO OBJECT-ORIENTED PROGRAMMING (OOP)

Basic Concepts

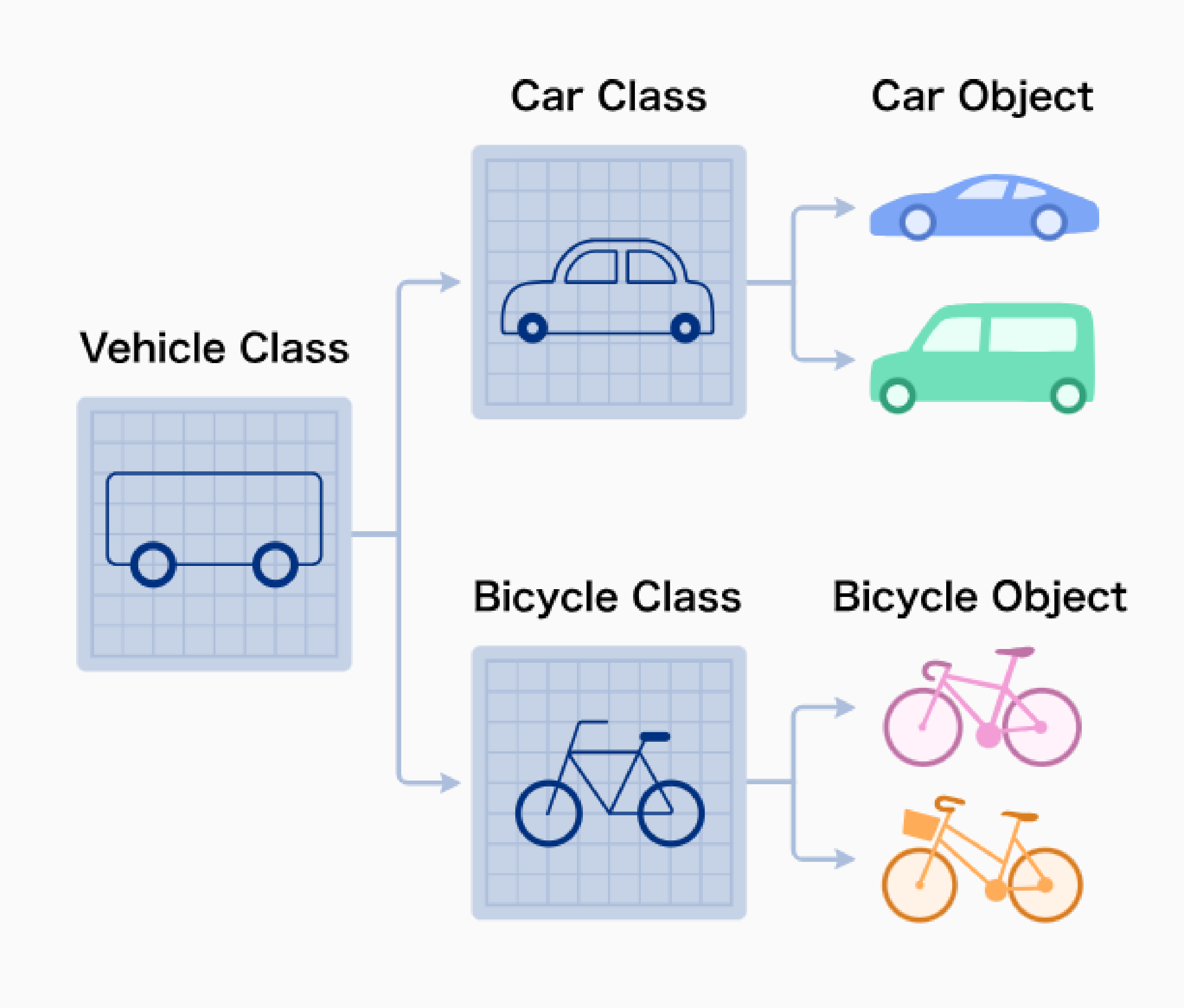
DEFINITION OF OOP

"Object Oriented Programming (OOP) is a programming methodology that links data structures (considered as "objects") with a set of operators (called "methods") that can act upon that data, essentially allowing for the representation of real-world entities and their behaviors within a program."

Object Oriented Programming is a style of programming!

MODELING REAL-WORLD ENTITIES THROUGH OBJECTS AND METHODS: EXAMPLE

Represent a vehicle with OOP



Methods

```
Move();
Stop();
```

```
Move();
Stop();
```

KEY PRINCIPLE OF OOP

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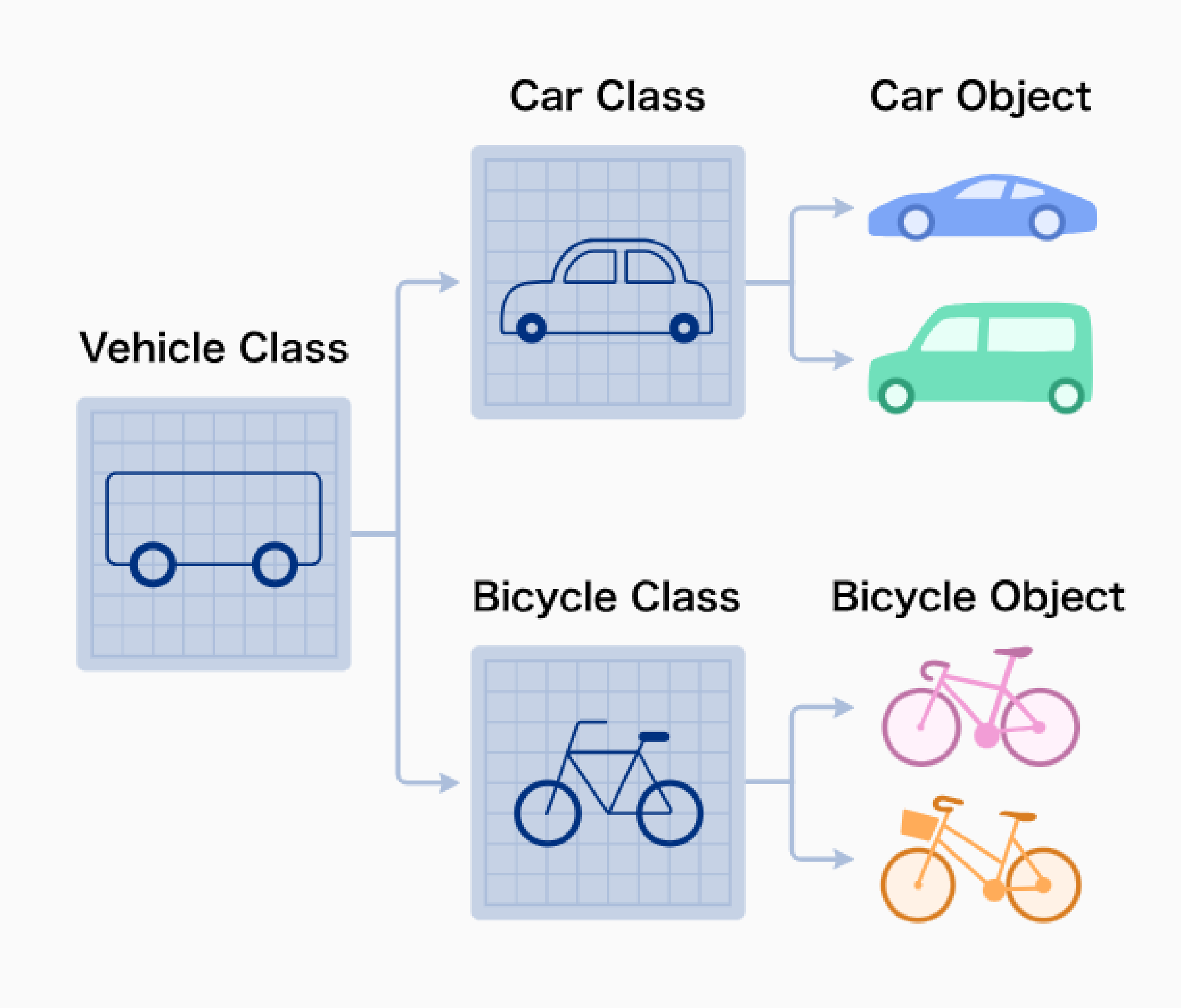
The paradigm of object-oriented programming is based on four main principles:

OOP PRINCIPLES

Encapsulation	Abstraction	Inheritance	Polymorphism
When an object only exposes the selected information.	Hides complex details to reduce complexity.	Entities can inherit attributes from other entities.	Entities can have more than one form.

These principles help in writing scalable, maintainable, and reusable code. OOP is usually applied for large-scale, complex programs that are actively updated or maintained.

ADVANTAGES OF OOP



Example:

- Encapsulation: Each class contains attributes and methods specific to it.
- Abstraction: The Vehicle Class is a generalized concept that defines common attributes and behaviors of all vehicles.
- Inheritance: Car and Bicycle inherit from the Vehicle Class.
- Polymorphism: The `move()` and `stop()` methods apply to both Car and Bicycle.

ADVANTAGES OF OOP

- Today's programs are massive, and writing programs gets harder as the program gets bigger and the team gets bigger.
- OOP tries to help to make easier for your programming to scale.

INTRODUCTION TO JAVA

Basic Concepts

What is Java?

WHAT IS JAVA?

- Developed in the early 90s by Sun Microsystems.
- High-level language, object-oriented programming (OOP).
- Portable across platforms
  - Each program is translated into Java bytecode.
  - Each machine has a Java Virtual Machine (JVM) which knows how to execute Java bytecode.

JAVA FEATURES

- Simple
- Platform independent
- Portable
- Dynamic and extensible
- Object-oriented
- Robust
- Secure
- Distributed
- Multi-threaded and interactive
- High performance

Further reading: https://techvidvan.com/tutorials/features-of-java-programming-language/

JAVA VIRTUAL MACHINE

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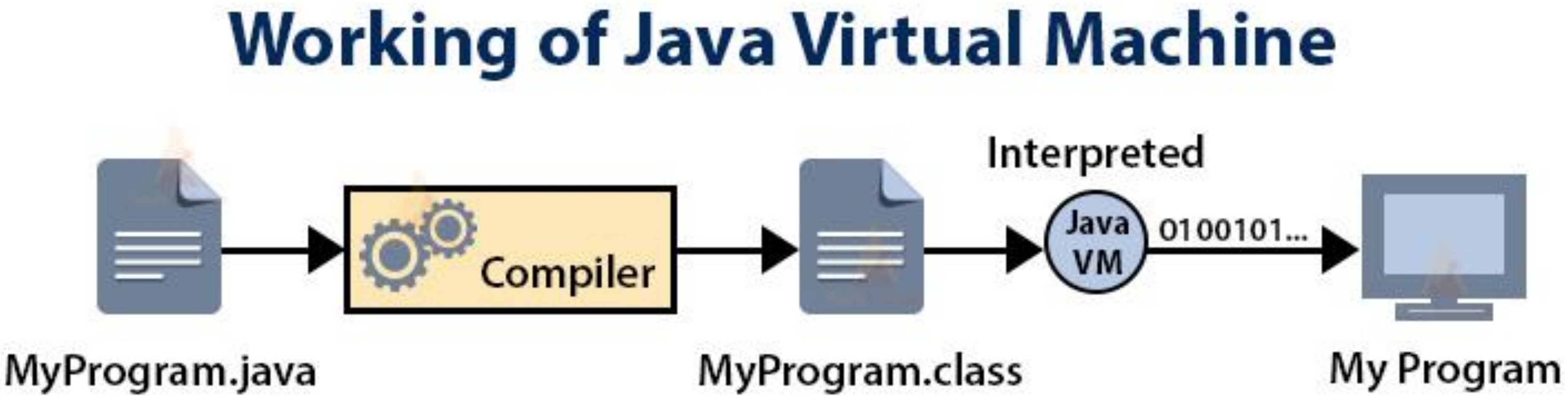
- Java compiler (javac) compiles the java source code into the bytecode.
- Java Virtual Machine (JVM) then executes this bytecode which is executable on many operating systems.



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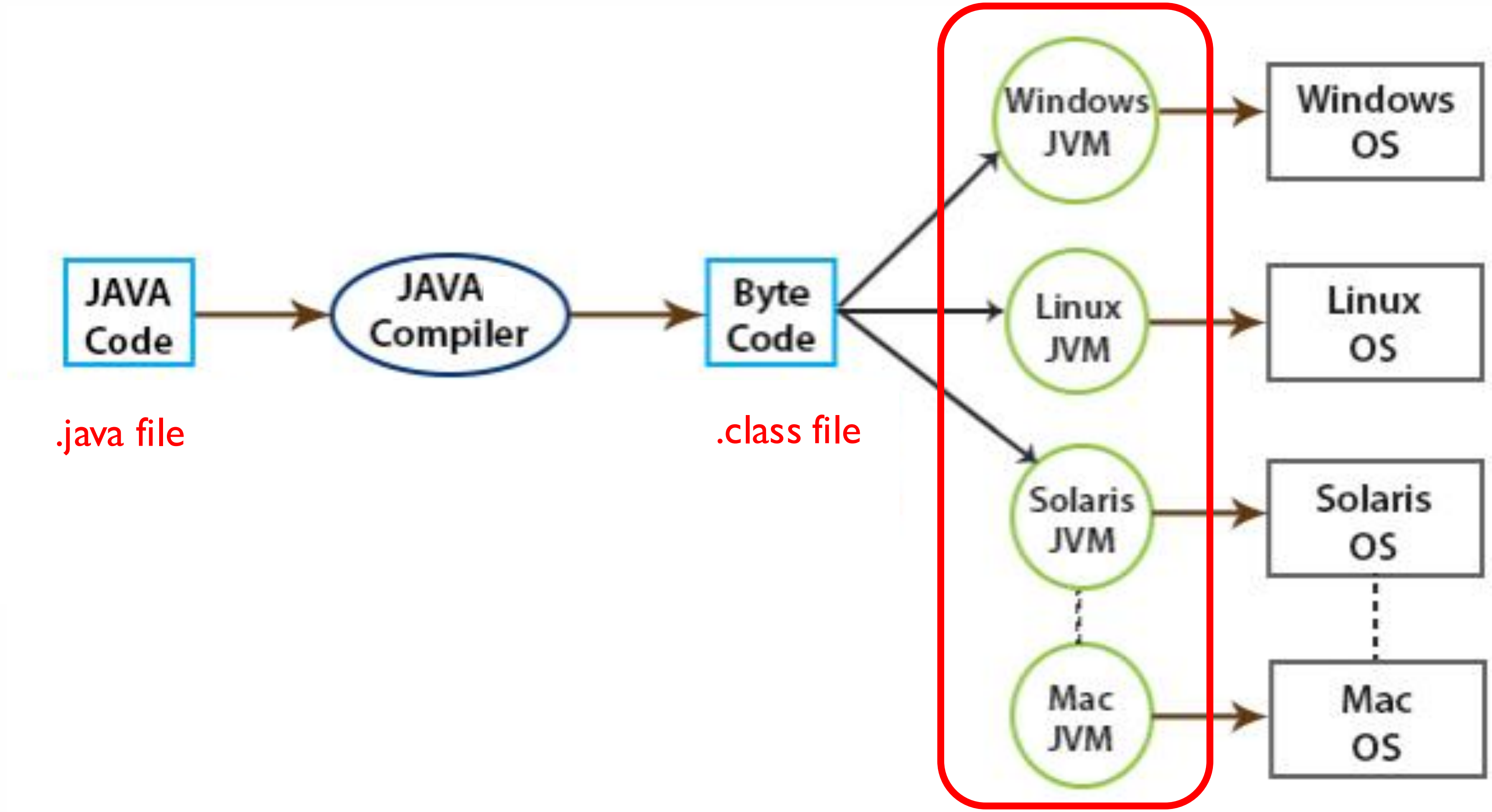
PLATFORM INDEPENDENCE IN JAVA

Java allows developers to write code that would be able to run on any machine or device, regardless of its architecture or platform.



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JAVA GARBAGE COLLECTION

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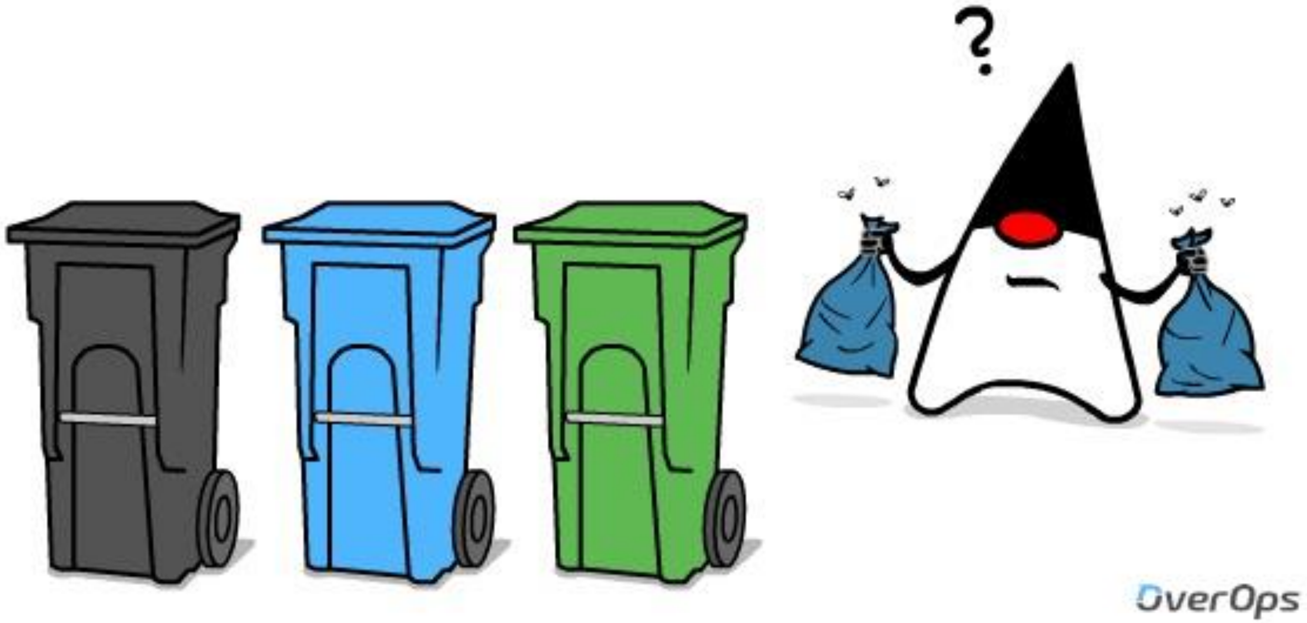
Java Virtual Machine (JVM) automatically deallocates the memory blocks and programmers do not have to worry about deleting the memory manually as in case of C/C++.



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Why is JAVA for OOP?

WHY IS JAVA FOR OOP?

- Java has a much cleaner, designed-in syntax for OOP.
- Java is almost purely object-oriented. You can not even write a “Hello World” program without creating a class!
- Other programming languages, such as Python does not require you to use classes and you can program in Python entirely as a procedural language.

THANK YOU!

Lecture 02: Variable and Operators